

Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 9

VTITLE: Implementation of V lab Weber's Law

Objective:

After performing this experiment

1. You will know the human ability of distinguishing differences in rectangular shapes.
2. You will know the human ability of distinguishing differences in circular shapes.
3. You will know the human ability of distinguishing differences in color (opacity).
4. You will be able to compute the Weber fraction (K) for various measures mentioned above.

Expected OUTCOME of Experiment: CO 2

- Students will understand and quantify perceptual sensitivity by observing the minimum noticeable differences in shape, size, and color as described by **Weber's Law**.
- Learners will be able to analyze and plot the relationship between the **percentage difference** and the **number of attempts**, demonstrating how perception accuracy improves with repeated trials.

Books/ Journals/ Websites referred:

- <https://cpe-iitg.vlabs.ac.in/exp/webers-law/procedure.html>

Background Theory:

Weber's Law

Understanding human ability of noticing minute differences in shapes, colors, heights etc can help graphic designers in designing better interfaces and interactions. Perceptual ability to make out differences in objects, colors, shapes, texts etc influences the cognition and the interaction efficiency.

Can we perceive the difference between a line of 10cm length against a line of 10.05 cm ? Can we perceive the difference in holding a weight of 10.0 kg against a weight of 10.05 kg ? This incremental threshold for detecting the difference in any observed value is an important human factor useful in designing GUIs.

For an initial stimulus value I (say 10 cm) let the increment threshold for detecting a difference be ΔI (say 0.05 cm).

Weber's Law (E. H. Weber, in 1834) states that the ratio of $\Delta I/I$ is constant for a specific measure.

where ΔI = Differential threshold I = Initial stimulus intensity K = Weber fraction

Weber's Law states that the size of the just noticeable difference (jnd) is a constant proportion(K times) of the original stimulus value. It is the minimum amount by which stimulus intensity must be changed in order to produce a noticeable variation in sensory experience.

Example

Let two circles filled with same color and intensity be displayed side by side. Suppose the initial color intensity is 100 for both circles. Now let us keep changing the color intensity of any one circle incrementally till the user perceives noticeable change or difference in their intensities. We will stop at this point and record this new color intensity. Let this new value be 110. Thus the differential threshold (jnd) = 10 ($\Delta I = 110 - 100$) and the Weber's constant in this case would be $K = 10/100 = 0.1$. If the Weber fraction for discriminating changes in stimulus is a constant proportion equal to 0.1 then the size of the just noticeable difference for a spot having an intensity of 1000 would be 100.

Some interesting human-sense JNDs (Just Noticeable Differences) are tabulated below :

Pitch: 1/333 Brightness: 1/60 Lifted Weights: 1/50

Loudness: 1/10 Pressure on skin: 1/7 Taste: 1/5

(Reference : http://www.richardbrice.net/webers_law.htm)

Notice how our sensitivity to pitch is 70 times more acute than our sense of taste.

Webers Law as applied to user interfaces

Weber's Law can be used for various sensory modalities in GUIs such as brightness, loudness, line length, visual weight of fonts in typography, color matching etc. Many times large amount of information is required to be displayed on a limited size computer screen. Information is displayed in various forms viz. text, pictures, drawings, maps, graphs, videos etc. Poor visual design of user interface lacks the ability of differentiating between two close enough visual stimuli e.g. two lines with different thicknesses in a map (for wide roads and narrow lanes). What is this threshold of line thickness that may lead to noticeable difference is governed by Weber's law. The threshold of noticeable difference between color shades is also governed by Weber's law.

Procedure

Experiment 1 (Changing Shape – Rectangle)

1. You will see 6 blue coloured rectangular blocks on the screen.
2. Your time will start after you press the 'START' button.
3. Observe all blocks minutely and identify the one that expands in breadth after some time
4. The moment you observe the difference click on the same block to stop the timer.
5. Record the time and % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% area difference' Vs 'number of attempts'
7. To repeat the same experiment first reset the timer by pressing the 'RESET' button.

Experiment 2 (Changing Shape - Circle)

1. You will see 6 black colored circles on the screen.
2. Your time will start after you press “START” button.
3. Observe all circles minutely and identify the one that expands in radius after some time.
4. The moment you observe the difference click on the same circle to stop the timer.
5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% radial difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing “RESET” button.

Experiment 3 (Changing Color - Circle)

1. You will see 6 black colored circles.
2. Your time will start after you press “START” button.
3. Observe all circles minutely and identify the one that changes color after some time.
4. The moment you observe the difference click on the same circle to stop the timer.
5. Record the time and the % noticed difference or the % color difference.
6. Repeat the step 1-5 and plot the graph between '% color difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing “RESET” button.

VLAB NAME : Weber’s Law

Part- 1 :

Note : use start only once at a time till you **reset**.

Instructions

1. You will see 6 green colored rectangular blocks on the screen.
2. Your time will start after you press "START" button.
3. Observe all blocks minutely and identify the one that expands in breadth after some time.
4. The moment you observe the difference click on the same block to stop the timer.
5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% area difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



5 s

$A = 34t + 5600$
 $A = 5770$
 $\therefore A..diff.. \% = 3.04$

---END OF EXPERIMENT---

Note : use start only once at a time till you **reset**.

Instructions

1. You will see 6 green colored rectangular blocks on the screen.
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3. Observe all blocks minutely and identify the one that expands in breadth after some time.
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5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% area difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



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 $A = 5770$
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---END OF EXPERIMENT---

Note : use start only once at a time till you reset.

Instructions

1. You will see 6 green colored rectangular blocks on the screen.
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5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% area difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



6 s

$A = 34t + 5600$
 $A = 5804$
..A..diff..%=3.64

---END OF EXPERIMENT---

Note : use start only once at a time till you reset.

Instructions

1. You will see 6 green colored rectangular blocks on the screen.
2. Your time will start after you press "START" button.
3. Observe all blocks minutely and identify the one that expands in breadth after some time.
4. The moment you observe the difference click on the same block to stop the timer.
5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% area difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



5 s

$A = 34t + 5600$
 $A = 5770$
..A..diff..%=3.04

---END OF EXPERIMENT---




Weber's Law



Rate Me
Report a Bug

Note : we start only once at a time till you **reset**.

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7. To repeat the same experiment first reset the timer by pressing "RESET" button.



start
reset



start
reset

7 s

$$A = 34t + 5600$$

$$A = 5838$$

$$\therefore A_{diff} \% = 4.25$$

---END OF EXPERIMENT---

Part-2 :

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☰  Virtual Labs

Spot expanding circle and double click

Note : It will not work for single click, as it will give wrong information

Instructions

1. You will see 6 blue colored circles on the screen.
2. Your time will start after you press "START" button.
3. Observe all circles minutely and identify the one that expands in radius after some time.
4. The moment you observe the difference click on the same circle to stop the timer.
5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% radial difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



start

reset

cpe-iitg.vlabs.ac.in says

$A = 34t + 5600 \pm A = 5770, A_{diff. \%} = 3.04$

OK

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 Virtual Labs

Spot expanding circle and double click

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start

reset

cpe-iitg.vlabs.ac.in says

$A = 34t + 5600$; $A = 5838$; Δ .diff.% = 4.25

OK

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 Virtual
LABS

Spot expanding circle and double click

Note : It will not work for single click, as it will give wrong information

Instructions

1. You will see 5 blue colored circles on the screen.
2. Your time will start after you press "START" button.
3. Observe all circles minutely and identify the one that expands in radius after some time.
4. The moment you observe the difference click on the same circle to stop the timer.
5. Record the time and the % noticed difference or the % area difference.
6. Repeat the steps 1-5 and plot the graph between '% radial difference' Vs 'number of attempts'.
7. To repeat the same experiment first reset the timer by pressing "RESET" button.



start

reset

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$A = 34t + 5600$; $A = 6008$; $A_{diff} \% = 7.29$

OK

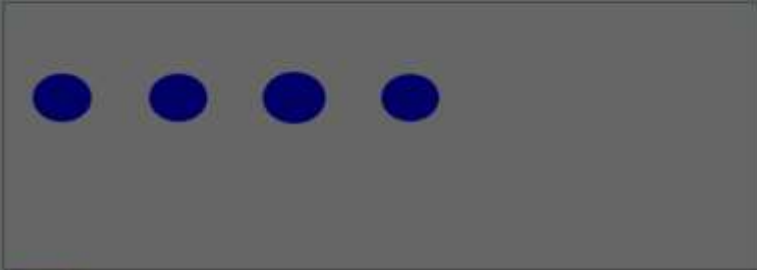
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Virtual Labs

Spot expanding circle and double click

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Instructions
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 5. Record the time and the % noticed difference or the % area difference.
 6. Repeat the steps 1-5 and plot the graph between "% radial difference" Vs "number of attempts".
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start
 reset

cpe-iltg.vlabs.ac.in says
 $A = 34t + 5600$; $A = 5838$, $A_{diff} \% = 4.25$

OK

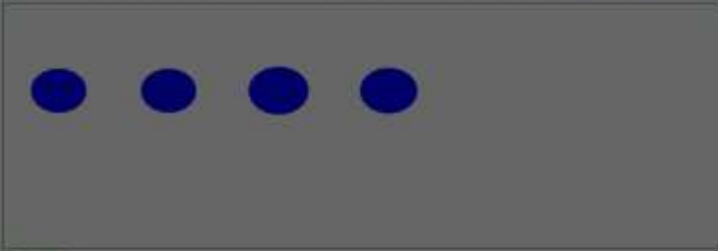
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Virtual Labs

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start
 reset

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 $A = 34t + 5600$; $A = 5770$, $A_{diff} \% = 3.04$

OK